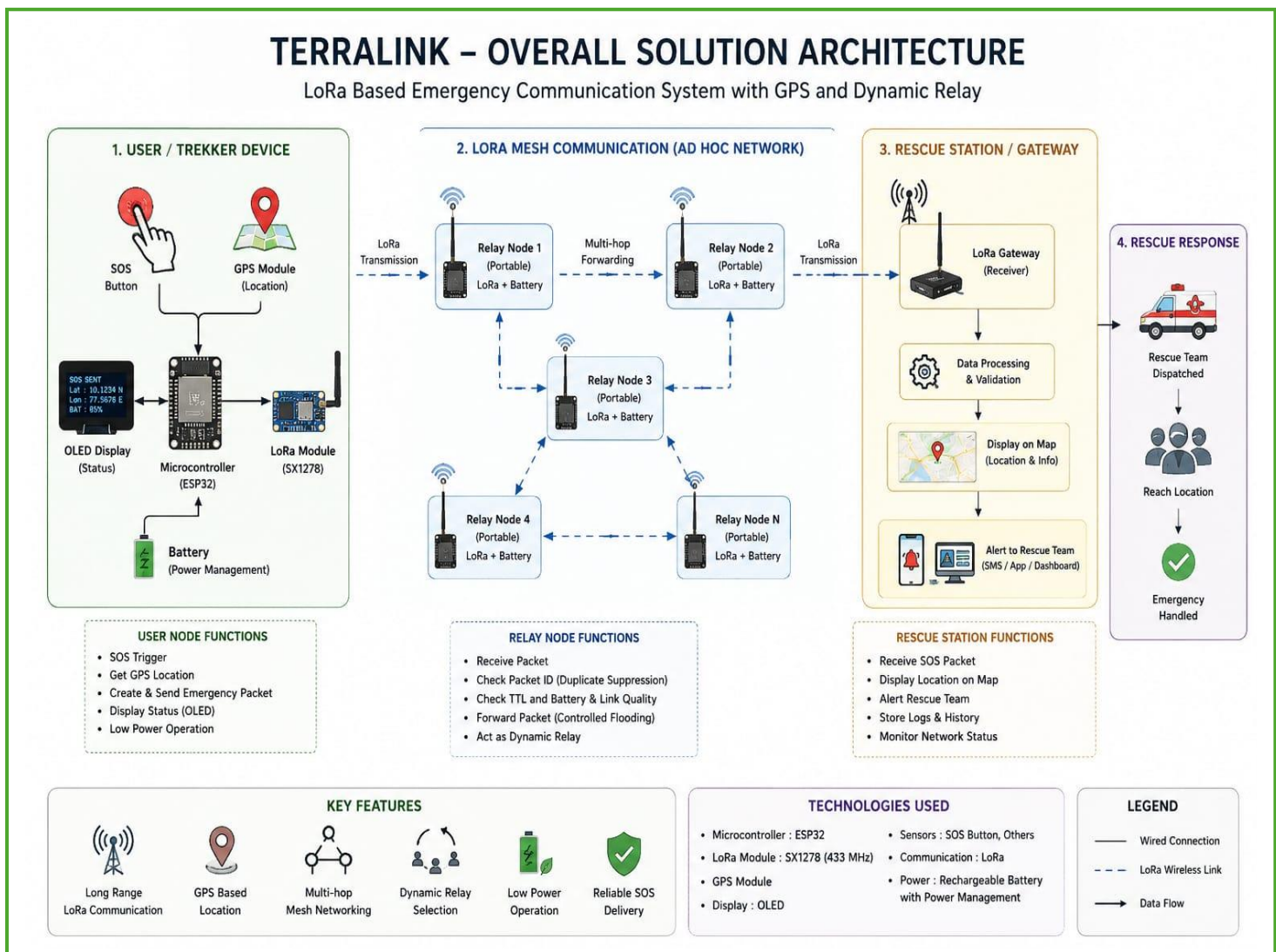


# TERRALINK

## SYSTEM ARCHITECTURE

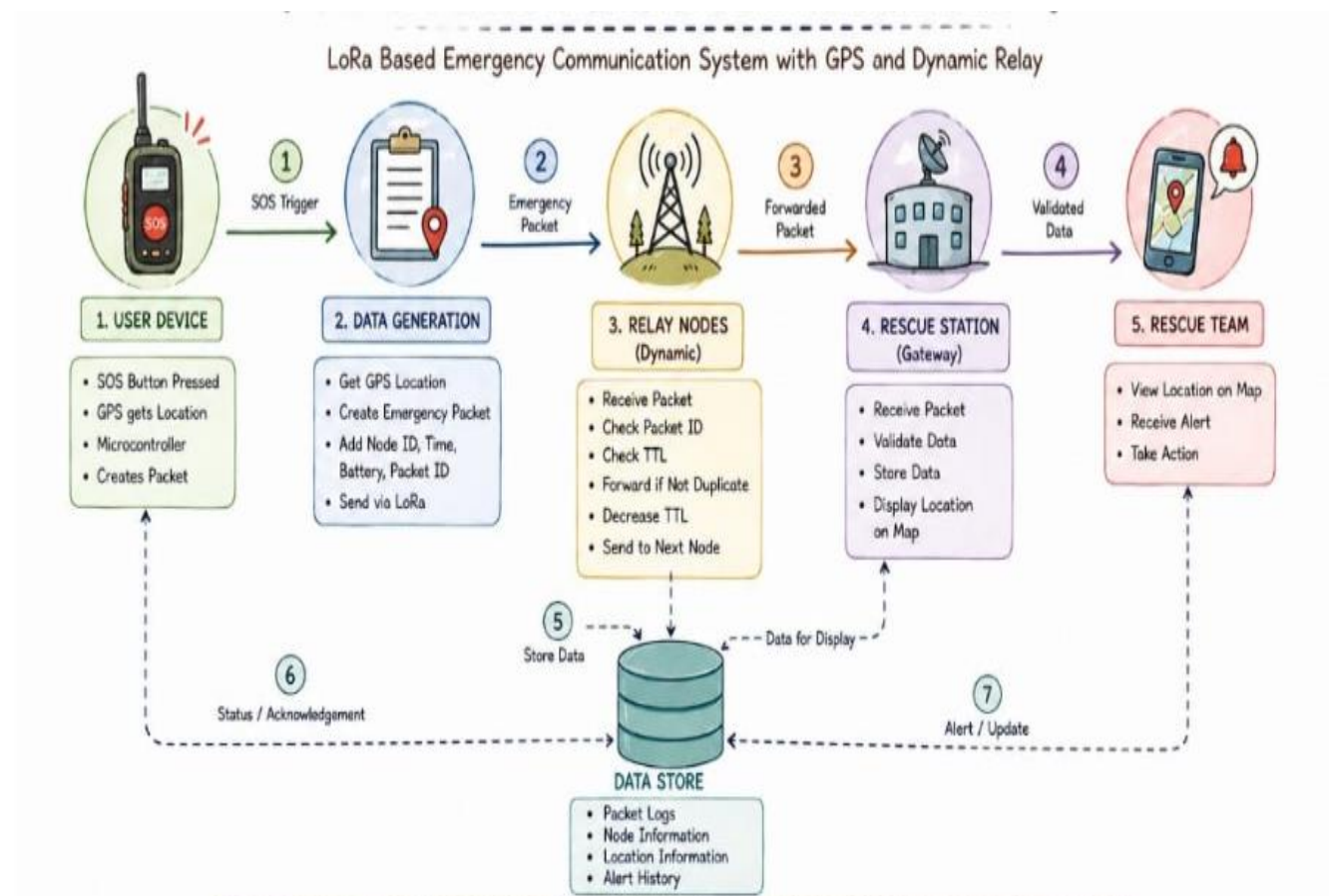
### Overall Solution Architecture:

TerraLink collects the emergency trigger and GPS location data through the touch sensor, GPS module, and microcontroller. The emergency data is converted into an SOS packet and transmitted through LoRa communication. Relay nodes forward the packet until it reaches the rescue station, where the emergency message and location are received for quick response.



## DATA FLOW DIAGRAM:

TerraLink collects the emergency trigger and GPS location data through the touch sensor, GPS module, and microcontroller. The emergency data is converted into an SOS packet and transmitted through LoRa communication. Relay nodes forward the packet until it reaches the rescue station, where the emergency message and location are received for quick response.



## HARDWARE SOFTWARE PARTITIONING:

In TerraLink, the system is divided into hardware and software components based on their functions. The hardware includes the microcontroller, LoRa module, GPS module, touch sensor, and OLED display, which handle sensing, communication, location tracking, and data display. The software

is responsible for processing sensor inputs, collecting GPS data, creating SOS packets, controlling LoRa communication, and displaying emergency information. This partitioning ensures efficient and reliable operation of the TerraLink emergency communication system.

| HARDWARE COMPONENTS           |                                                                                     |                                                                    | SOFTWARE MODULES              |                                                                                       |                                                                                                                                                |
|-------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Component                     | Image                                                                               | Function                                                           | Module                        | Image / Icon                                                                          | Function                                                                                                                                       |
| Microcontroller (ESP32)       |    | Controls all operations, processes data and manages communication. | 1. Device Management          |    | <ul style="list-style-type: none"> <li>Initialize hardware</li> <li>Device ID management</li> <li>Battery monitoring</li> </ul>                |
| LoRa Module (SX1278)          |    | Provides long range wireless communication.                        | 2. Sensor Management          |    | <ul style="list-style-type: none"> <li>Read touch sensor</li> <li>Detect SOS trigger</li> <li>Trigger events</li> </ul>                        |
| GPS Module (NEO-6M)           |   | Provides location information (latitude, longitude).               | 3. GPS Management             |   | <ul style="list-style-type: none"> <li>Acquire location</li> <li>Parse GPS data</li> <li>Provide coordinates</li> </ul>                        |
| Touch Sensor (TTP223)         |  | User triggers SOS alert using touch.                               | 4. Communication (LoRa Stack) |  | <ul style="list-style-type: none"> <li>Create packet</li> <li>LoRa transmission</li> <li>Packet reception</li> </ul>                           |
| OLED Display (0.96" I2C)      |  | Displays status, GPS location and alerts.                          | 5. Relay Management           |  | <ul style="list-style-type: none"> <li>Check packet ID</li> <li>TTL handling</li> <li>Forward packet</li> <li>Duplicate suppression</li> </ul> |
| Power Supply (Li-ion Battery) |  | Provides power to the entire system.                               | 6. Data Management            |  | <ul style="list-style-type: none"> <li>Store logs</li> <li>Store location</li> <li>Store alerts</li> </ul>                                     |
| Antenna (433 MHz)             |  | Enhances LoRa signal transmission.                                 | 7. User Interface             |  | <ul style="list-style-type: none"> <li>Display status on OLED</li> <li>Show location</li> <li>Show alerts</li> </ul>                           |
|                               |                                                                                     |                                                                    | 8. Power Management           |  | <ul style="list-style-type: none"> <li>Sleep / Deep sleep modes</li> <li>LoRa power control</li> <li>Optimize power usage</li> </ul>           |

## FUNTIONAL BLOCK DIAGRAM:

**Emergency Detection:** Touch sensor detects an emergency.

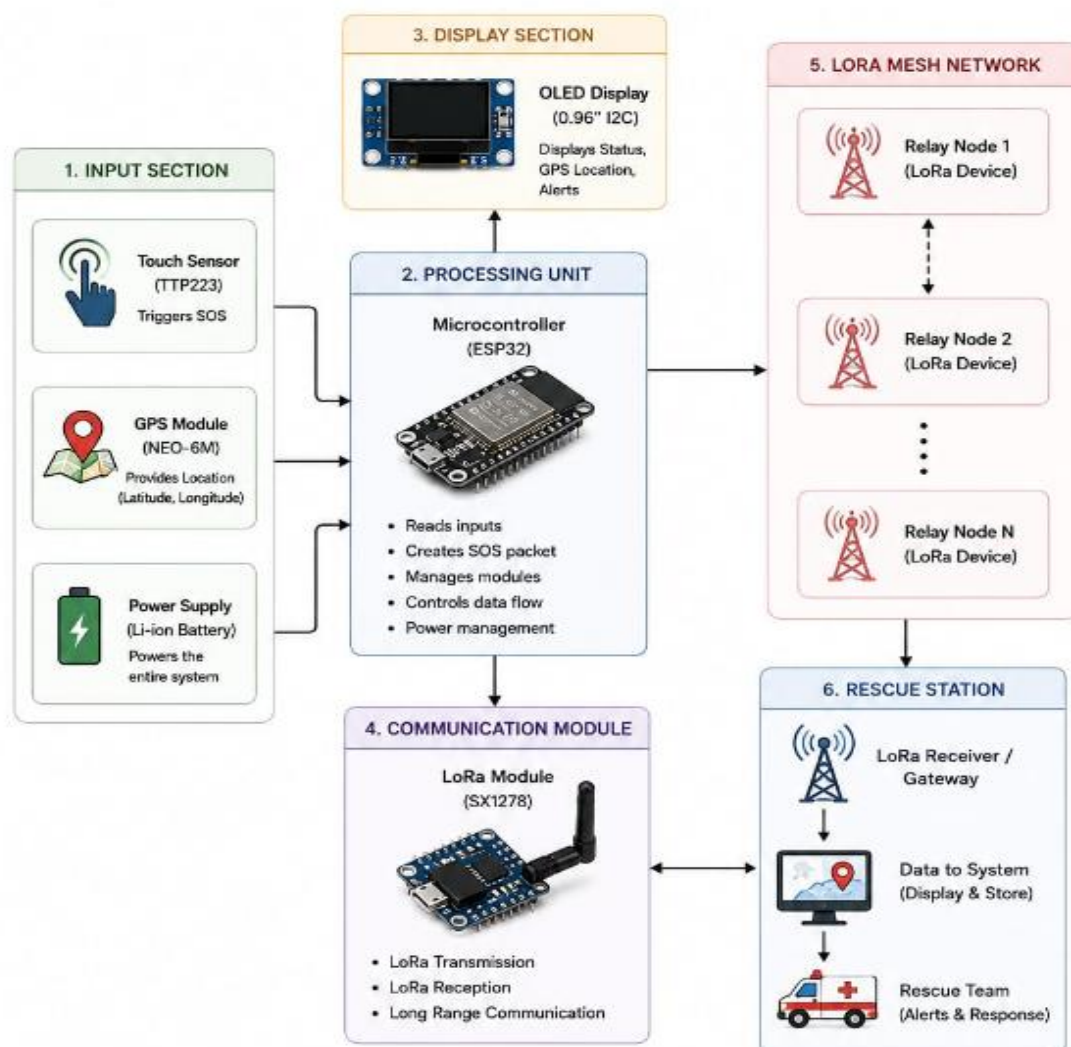
**Data Processing:** Microcontroller processes the emergency signal.

**Location Tracking:** GPS obtains the user's location.

**LoRa Transmission:** Emergency message and location are transmitted through LoRa.

**Relay Nodes:** Relay nodes forward the message to the destination.

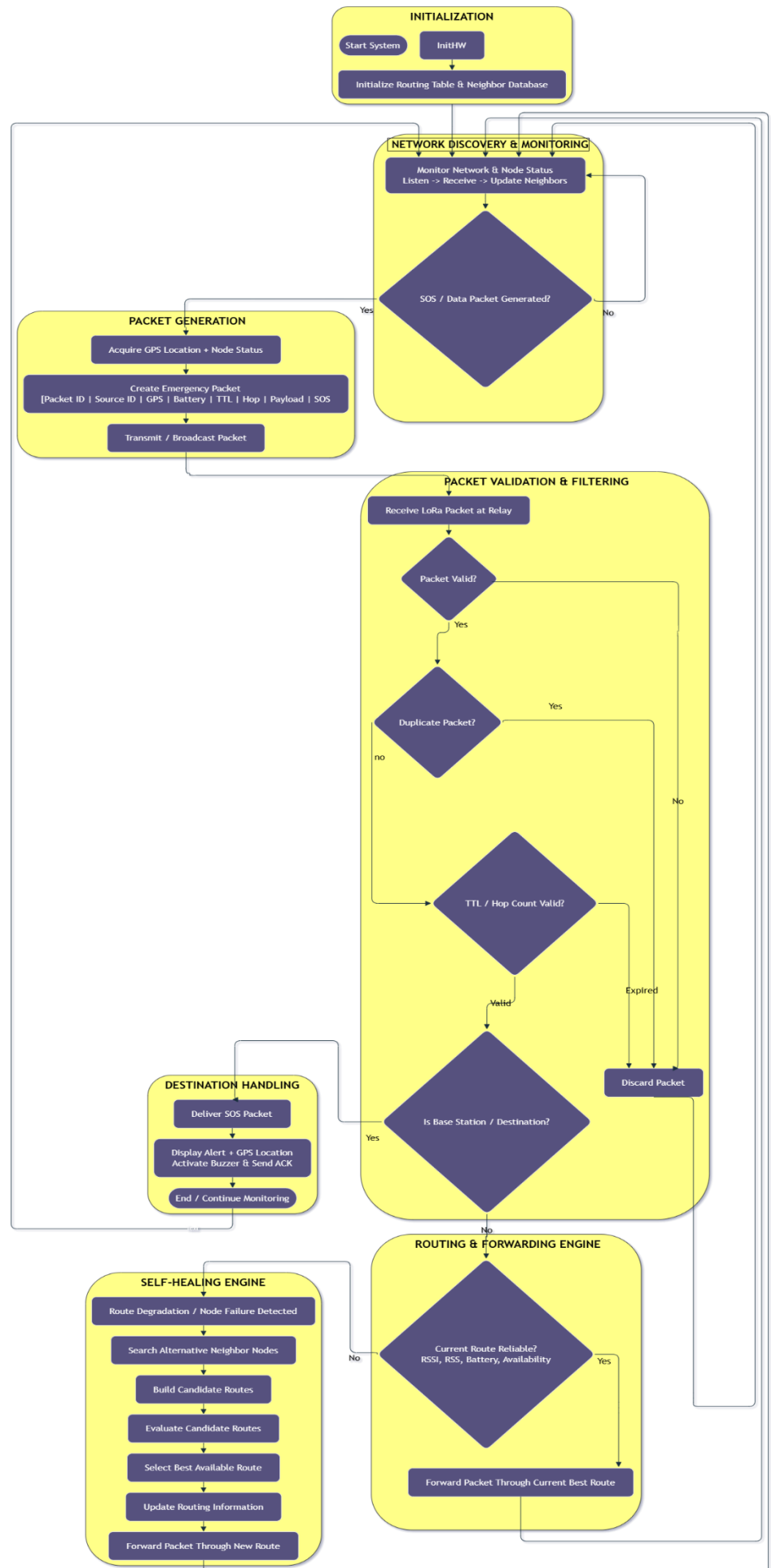
**OLED Display:** OLED displays system status and location information.







# FLOW CHART:

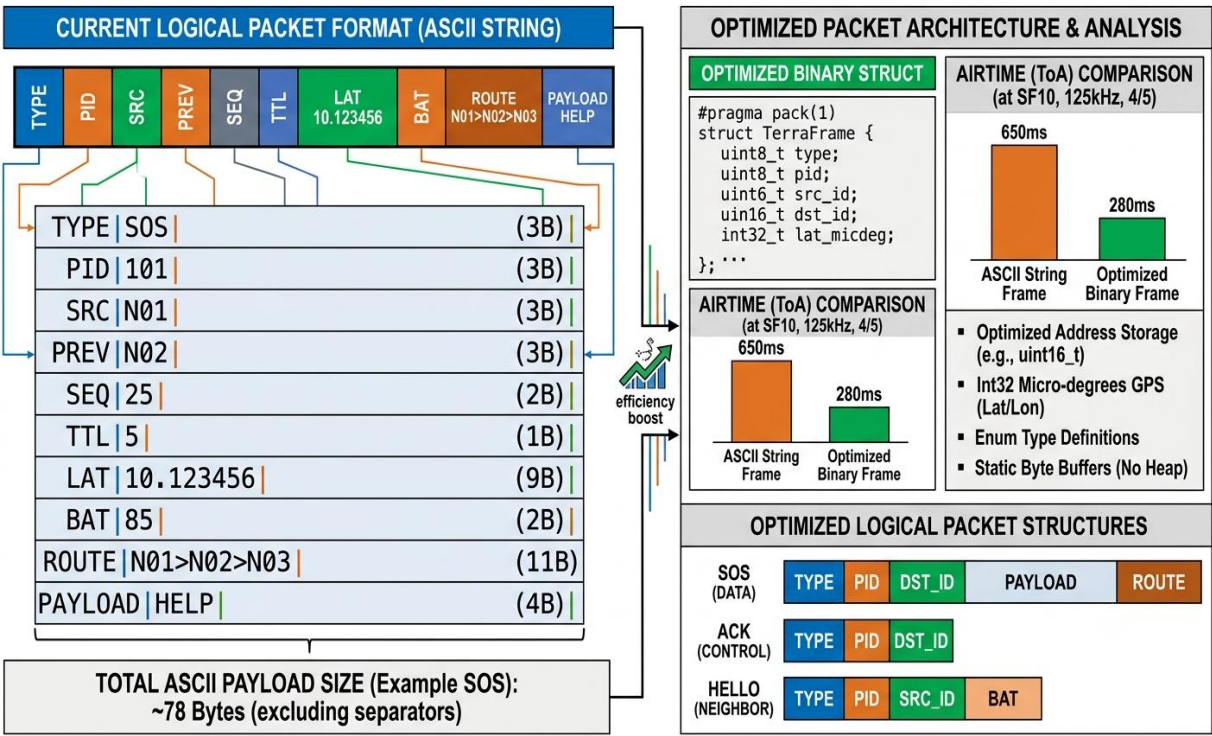


# COMMUNICATION PROTOCOL

The Dynamic TerraLink Communication Protocol defines how an SOS packet is **created, routed, validated, forwarded, acknowledged, and recovered when a communication link fails.** It uses lightweight AODV-inspired routing, duplicate filtering, TTL-based loop prevention, hop-by-hop ACK, retransmission, and self-healing.

## 1. Packet Creation and Packet Structure

When an End Node detects an SOS event, it creates a structured packet containing the source, destination, sequence number, TTL, location, battery status, and emergency payload.



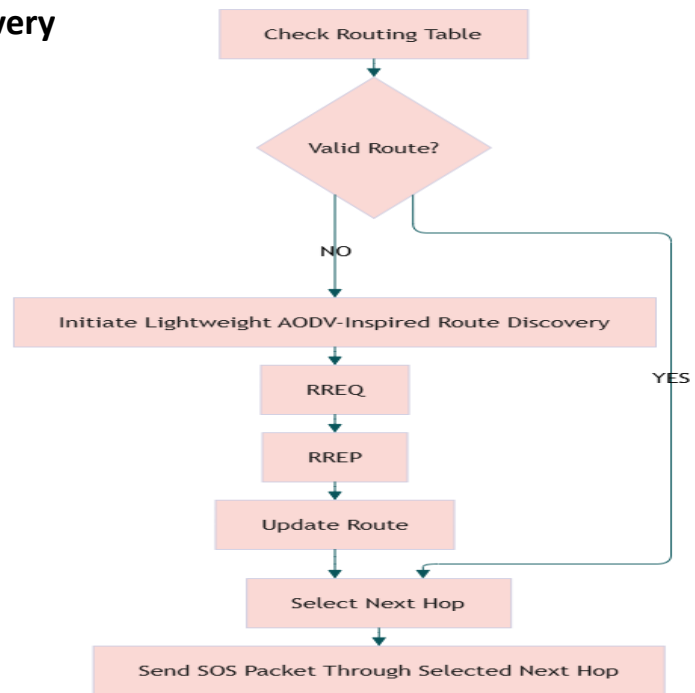
The main fields are:

- **TYPE** – Identifies the packet type.
- **PID** – Packet identifier.
- **SRC** – Source node.
- **DST** – Destination/base station.
- **SEQ** – Sequence number used for packet identification and duplicate filtering.
- **TTL** – Limits packet lifetime.
- **GPS** – Emergency location.
- **BATTERY** – Source-node battery status.
- **PAYLOAD** – Emergency information.

## 2. Route Selection and Route Discovery

Before transmitting an SOS packet, the node checks its routing table.

TerraLink uses a **lightweight AODV-inspired route discovery mechanism** rather than claiming full standard AODV implementation.

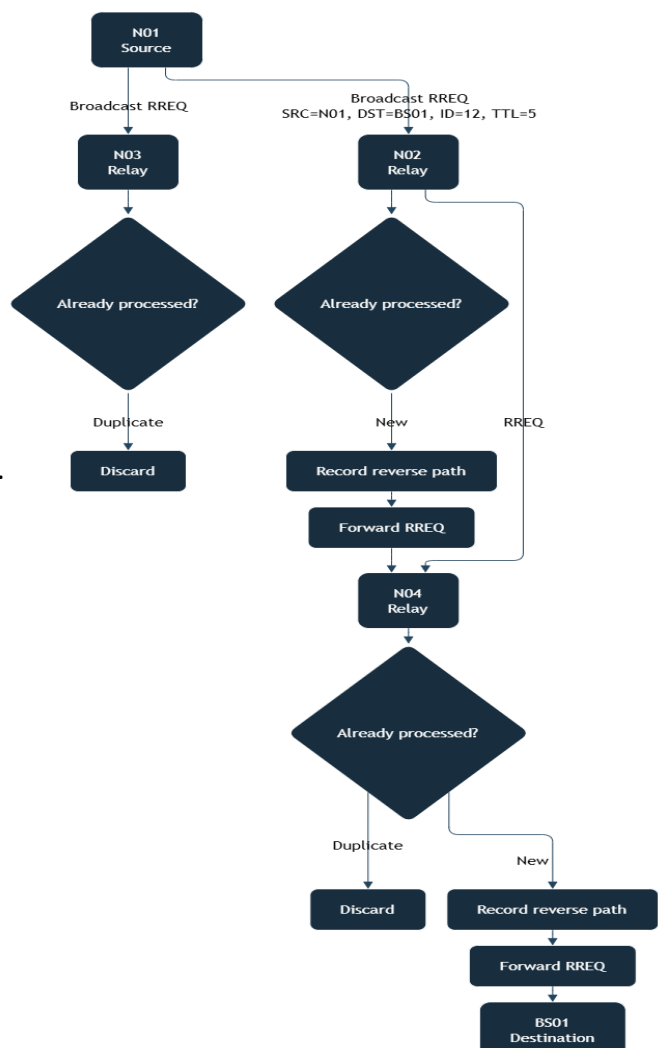


## 3. RREQ — Route Request

When there is no valid route, the source broadcasts an **RREQ**.

A relay receiving the RREQ:

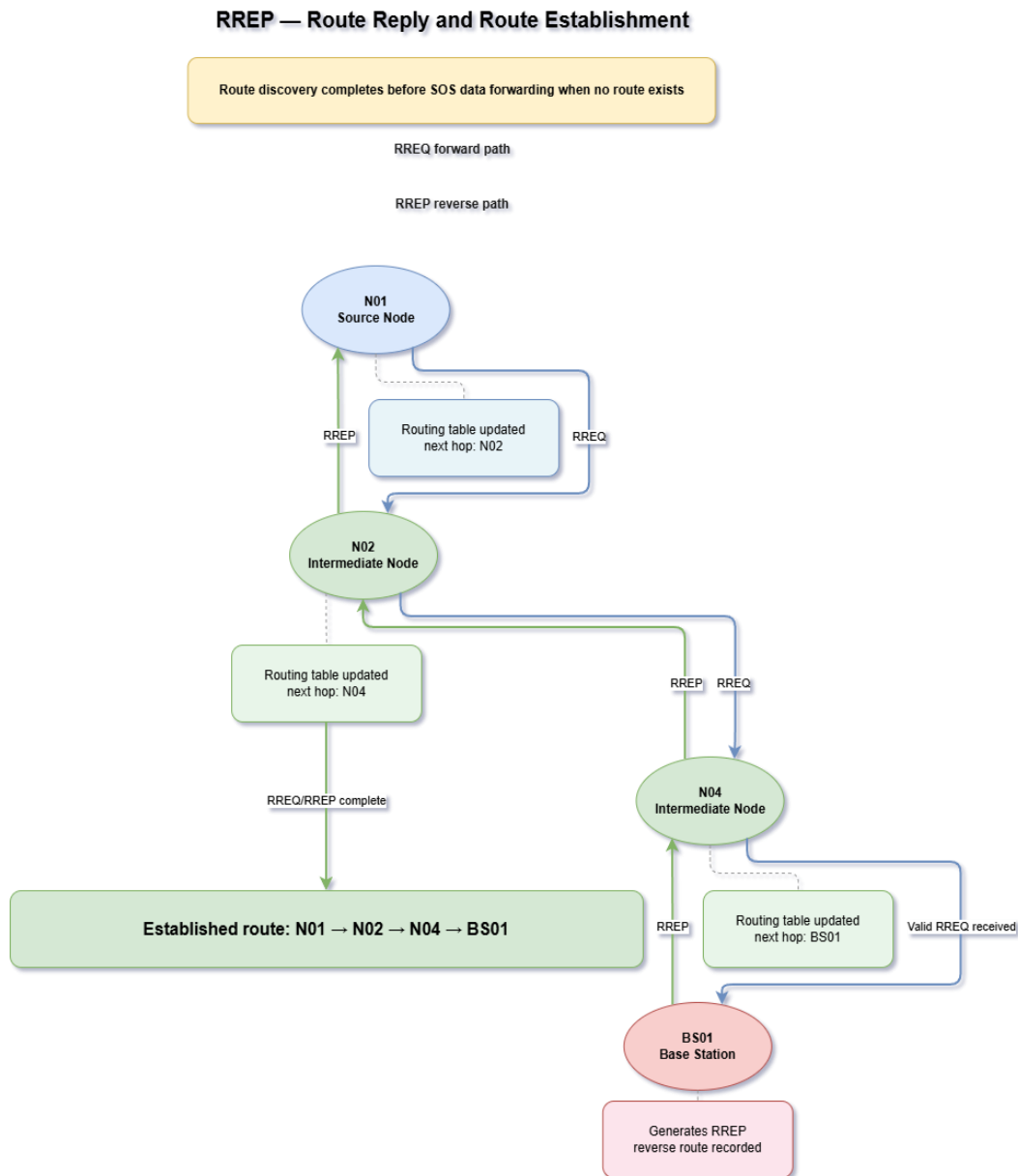
1. Checks whether this RREQ has already been processed.
2. If duplicate → discard.
3. If new → record the reverse path.
4. Forward the RREQ toward the destination.





#### 4. RREP — Route Reply

When the Base Station receives a valid RREQ, it generates an RREP. The RREP travels back through the reverse path:



The important point is that **RREQ/RREP establishes the route before the SOS data is forwarded when no route exists.**

#### 5. Duplicate Filtering

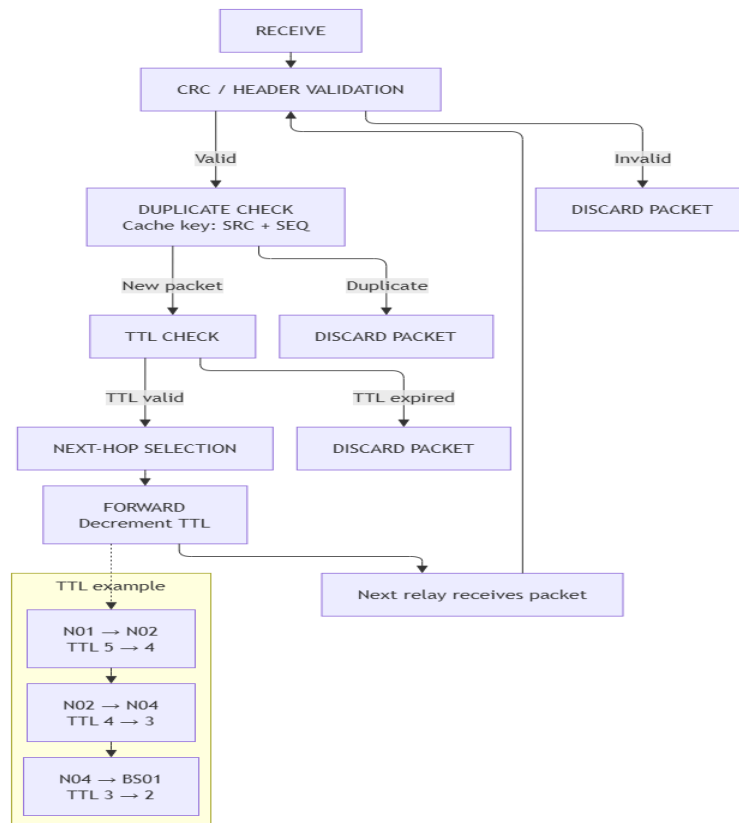
Each node maintains a small SRAM cache using:

**SRC + SEQ**

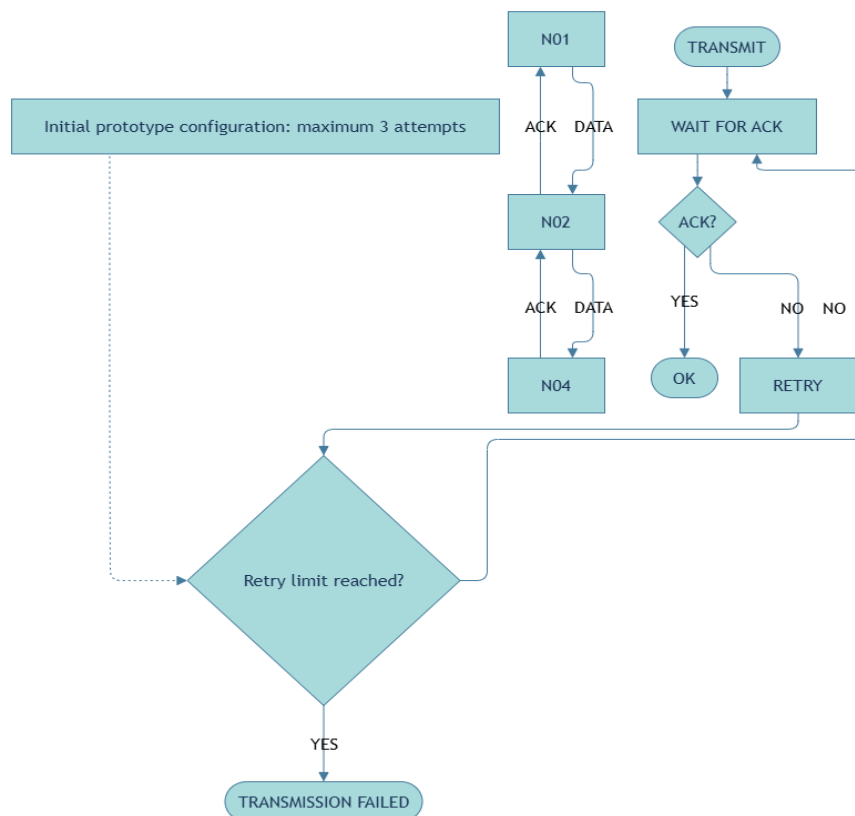
If the same source/sequence combination has already been processed, the packet is discarded to prevent repeated forwarding.

## 6. Packet Validation and Forwarding

Every relay validates a received packet before forwarding it. The processing order is:



## 7. Hop-by-Hop ACK and Retransmission

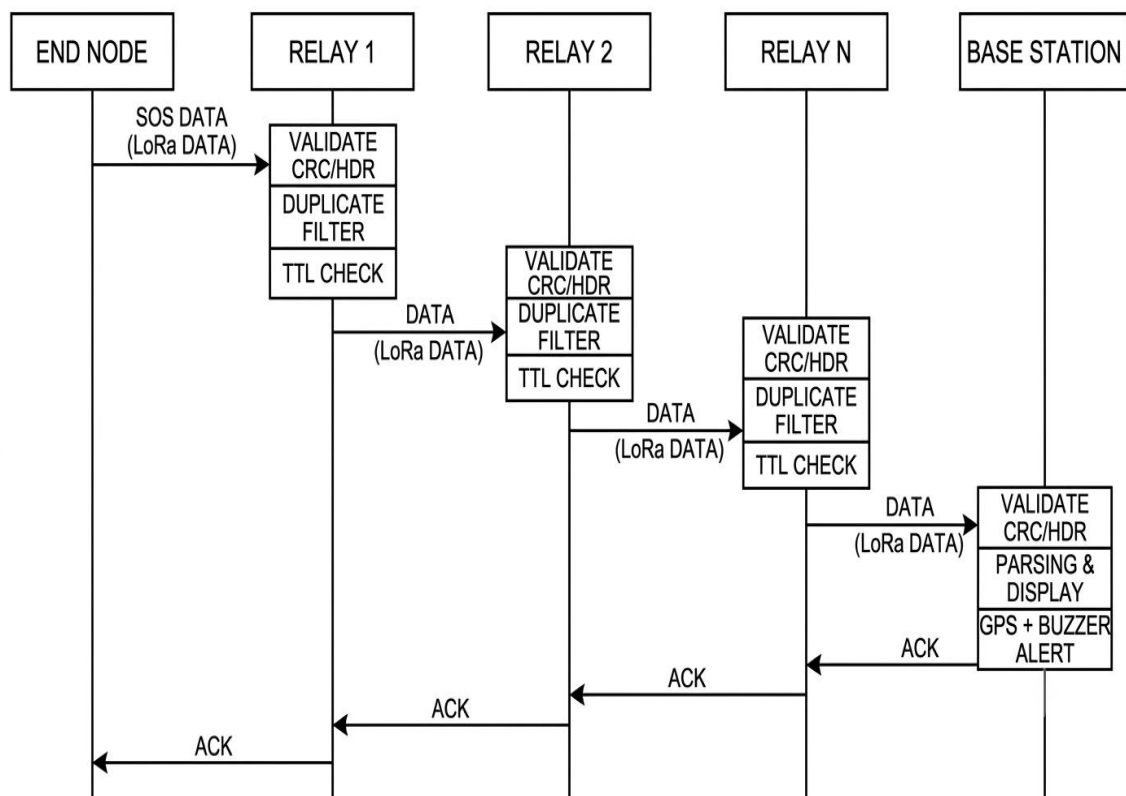
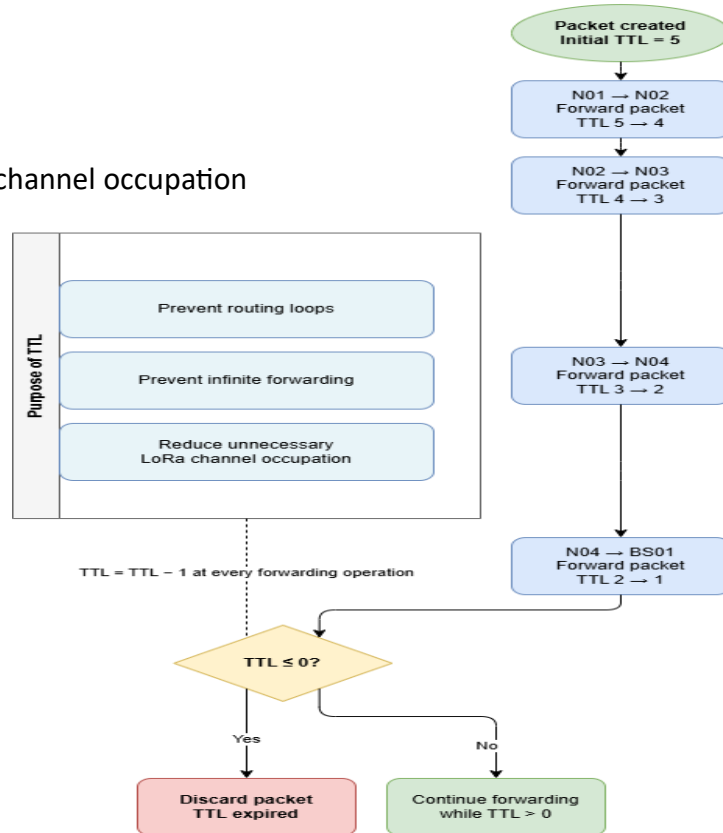


## 8. TTL — Packet Lifetime

TTL limits how many forwarding operations a packet can undergo.

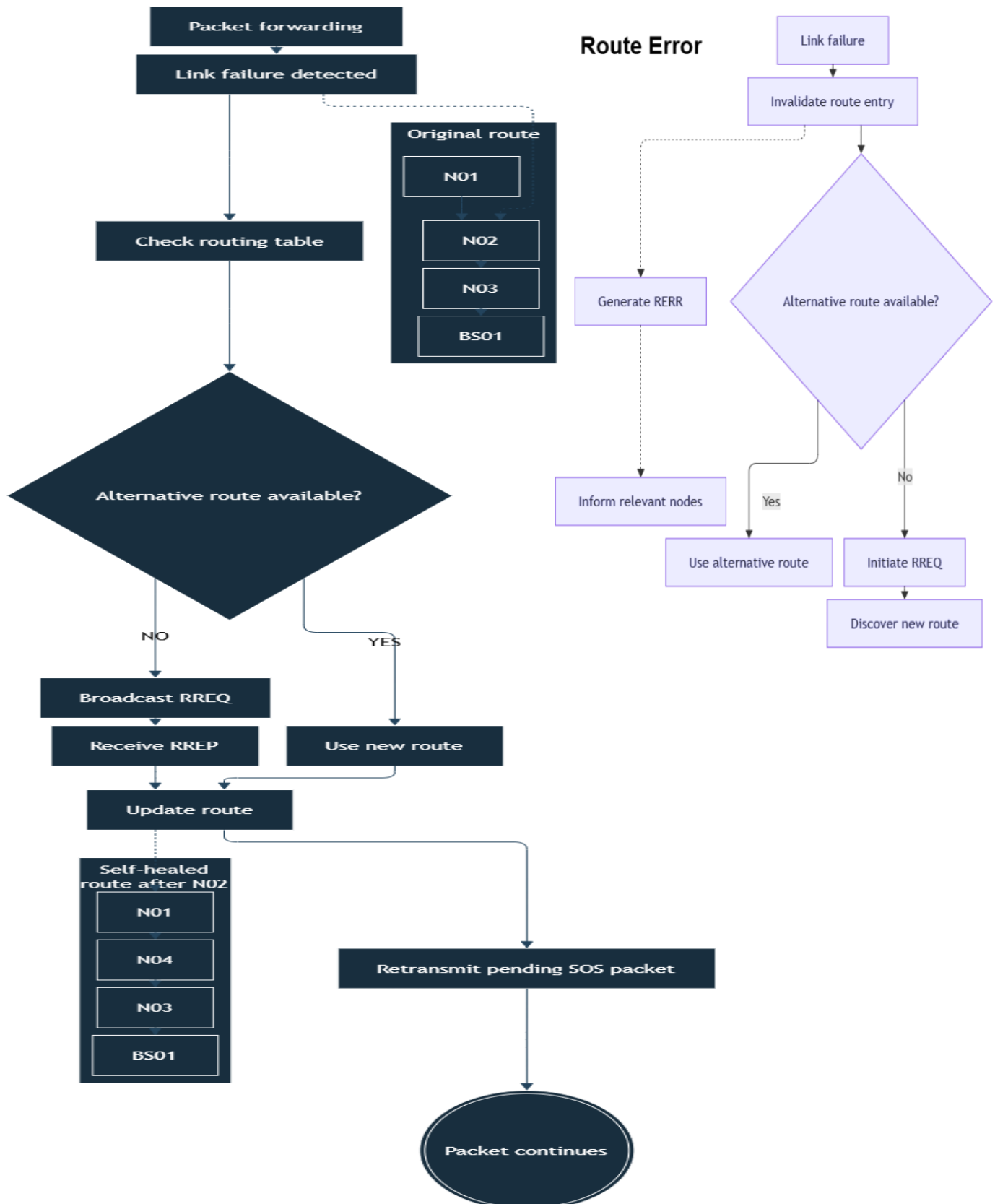
The purpose is to prevent:

- Routing loops
- Infinite forwarding
- Unnecessary LoRa channel occupation



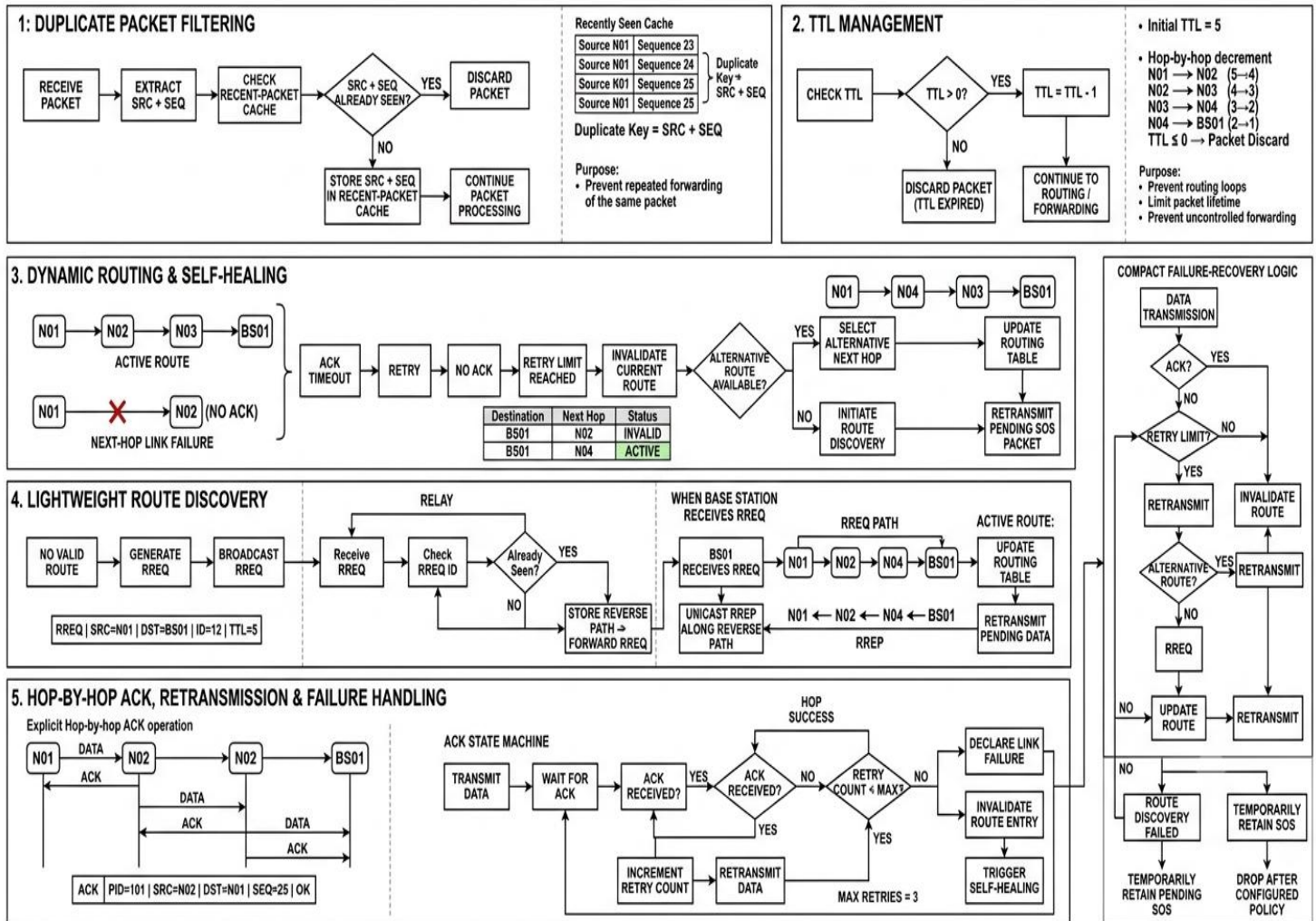
## 9. Self-Healing / Alternative Route

After detecting a failed next hop, the node checks whether another route is available.



# DYNAMIC TERRALINK — COMMUNICATION PROTOCOL: ROUTING, TTL, ACK & SELF-HEALING

LoRa-Based Multi-Hop Emergency Communication



The TerraLink protocol has **five core mechanisms**:

- 1. Routing:** Find and select the next hop.
- 2. Validation:** Reject corrupted, duplicate, or expired packets.
- 3. Forwarding:** Move the packet hop-by-hop toward the Base Station.
- 4. Reliability:** Use ACK + limited retransmission.
- 5. Self-healing:** Detect failed links, discover an alternative route, update the routing table, and retransmit the pending SOS.

TerraLink combines **LoRa multi-hop forwarding**, **SRC+SEQ duplicate filtering**, **TTL-based loop prevention**, **hop-by-hop ACK/retransmission**, and **lightweight AODV-inspired route recovery** to provide resilient emergency-message delivery.



### **BILL SUMMARY**

**Project Title :** TERRALINK

Certified that the following expenditures are made by us and the bills are admitted for the above project only. Bills are in order. It is assured that, no other claims are made in these bills.

| <b>Sl.No.</b> | <b>Bill No.</b> | <b>Bill Date</b> | <b>Description</b>                                                       | <b>Amount</b> |
|---------------|-----------------|------------------|--------------------------------------------------------------------------|---------------|
| 1.            | MZERD26-27/1930 | 07-05-2026       | Big Push Button Switch, AMS1117-3.3V, MT3608 Boost Converter             | ₹ 228.00      |
| 2.            | MZERD26-27/1930 | 07-05-2026       | TP4056 Module, 3.7V Li-ion Battery, NEO-6M GPS, AI Thinker LoRa SX1278   | ₹ 2696.00     |
| 3.            | MZERD26-27/1930 | 07-05-2026       | 18650 Single Cell Holder                                                 | ₹ 117.00      |
| 4.            | MZERD26-27/1930 | 07-05-2026       | ESP8266 CP2102, Breadboard, 4.5V Mini Solar Panel                        | ₹ 669.00      |
| 5.            | MZERD26-27/1930 | 13-05-2026       | TTP223 Touch Module, ARDUINO UNO R3, Wireless Antenna, 16×2 LCD with I2C | ₹ 999.00      |
| 6.            | MZERD26-27/1930 | 13-05-2026       | Breadboard GL-12                                                         | ₹ 91.50       |
| 7.            | MZERD26-27/1930 | 28-05-2026       | Arduino UNO R3                                                           | ₹ 496.50      |
| 8.            | MZERD26-27/1930 | 21-07-2026       | 0.96-inch OLED Display                                                   | ₹ 284.00      |
| TOTAL         |                 |                  |                                                                          | ₹ 5581.00     |

(Rupees FIVE THOUSAND FIVE HUNDRED AND EIGHTY ONE Only)